EEG Introduction and Project Overview:

Electroencephalography (EEG) records changes in electric potential over time through electrodes placed on the scalp. These recordings can help distinguish conditions or activities based on the signals themselves, however the biggest difficulty faced is noise. Between people EEG is extremely variable and is extremely tricky to actually connect to a certain pattern. Training a separate model from scratch for each subtask we want to perform with EEG is extremely tedious, which is what motivated the creation of **LaBraM and CBraMod**. These two models address this issue through the use of self-supervised pretraining, **learning numerical representations from EEG without requiring task-specific labels**. By crossing this barrier of pure brain wave recordings to embedding based representations, we are now able to innovate on the use of EEG with modern AI techniques and labeled examples. For this project, I investigated **whether pretrained features from CBraMod and LaBraM can help a classifier model distinguish whether a person's eyes are open or closed** based on a dataset of EEG recordings.

Introduction of CBraMod and LaBraM:

CBraMod divides each EEG channel into **short time segments called patches**, encoding each patch using two branches by using spatial and temporal attention. Spatial attention connects these patches from different electrodes during the same time interval while the temporal attention connects patches from different intervals within the same electrode channel. During self-supervised pretraining, some patches are hidden, and the **model learns to reconstruct their waveform by minimizing squared prediction error**. LaBraM also divides EEG channels into patches and trains a tokenizer that **assigns patch representations to entries in a learned codebook**, using reconstruction of spectral amplitude and phase to help the codes retain useful signal information. During pretraining, some of those patches are hidden and then the model learns to predict the codebook IDs based on the information from the existing patches. Effectively, **LaBraM builds a vocabulary of recording patterns** and then learns to identify which pattern belongs in a missing section, while **CBraMod learns to predict missing waveform values** based on the connection of space and time between patches.

Project Implementation:

To reiterate the project's goal before diving into the specific implementation and procedure, **our goal was to investigate whether pretrained features from CBraMod and LaBraM create a true difference in a classifier model** predicting whether a person's eyes are open or closed based on EEG recordings.

Dataset and Signal Preparation:

Using the recordings from twenty participants in the public **PhysioNet EEG Motor Movement/Imagery dataset**, I selected 19 electrode channels and applied common-average reference, subtracting their average voltage at each time point to establish a shared reference for the voltage measurements.

After applying this common average inference, I **filtered the EEG recordings to reduce very slow drift and fluctuations above 40 Hz** (keeping the 1-40 Hz range) for analysis. I then converted the recordings from 160 to 200 data points per second to match the input format of CBraMod and LaBraM. These additional points were estimated from the existing recording rather than newly measured. After removing the first and last two seconds as a buffer against disturbances at the recording boundaries, I divided the remaining signal into separate four-second clips. **Each clip became one classification example.** Finally as a method of filtration, I discarded any clip containing a voltage above +500 or below -500 μV to screen out extreme values.

Interpreting the Recordings and Training the Classifier:

Each recording window contained 19 channels with 800 measurements per channel. This calculation comes from dividing each channel into one-second patches with 4 patches per channel and 200 measurements per patch (19 x (4 * 200)). With every encoder transforming these patches into contextual representations, I was able to average the representations across channels and time patches (mean pooling), I was able to obtain a simple summary of our data.

To train, I kept the **encoders frozen and trained logistic regression on their features**. This procedure, called a linear probe, updates only the classifier's parameters using the eyes-open and eyes-closed labels. This then tests whether the underlying representations expose information useful for a classifier to use.

Comparing each pretrained encoder with a randomly initialized version of the same architecture, I was able to establish a baseline model where it would still learn to classify just not from the specialized embeddings created by the custom embedding models like CBraMod and LaBraM.

Evaluation and Results

I used three repetitions of five-fold participant-level cross-validation. Each outer fold held out four participants for testing, while four inner folds selected classifier regularization using only the remaining sixteen participants. Feature scaling was fitted within each training fold, and all windows from a participant stayed together to prevent their recordings from appearing in both training and testing. I calculated metrics separately for each participant and then averaged

them. Balanced accuracy measures the average recall across the two classes, while AUROC measures how well scores rank eyes-closed above eyes-open windows within a participant.

Representation	Balanced accuracy	Repeat SD (percentage points)	Mean participant AUROC
Bandpower	75.5%	3.3	**0.976**
CBraMod pretrained	77.9%	2.1	0.940
CBraMod random	77.8%	0.5	0.914
LaBraM pretrained	**78.7%**	1.1	0.947
LaBraM random	75.0%	1.2	0.892
Bandpower label control	44.1%	2.7	0.383

Based on the results, we see that pretrained LaBraM exceeded its random encode by 3.76 percentage points, with improvements for 12 of the 20 participants. CBraMod's corresponding gain was 0.11 points, **indicating little balanced-accuracy benefit from pretraining in this setup**. These differences were calculated from participant-level results before rounding. Bandpower achieved the strongest within-participant AUROC despite lower balanced accuracy, suggesting that useful ranking information did not always translate into correct decisions for the classifier. Noise due to the variation in participants could be one reason for this problem.